

CIDEX — The Compression-Ignition Engine Certification Panel

Harmonization pipeline, validation, and provenance

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Abstract

The United States Environmental Protection Agency publishes emission certification records for compression-ignition engines as two Microsoft Excel workbooks whose internal structures are mutually incompatible. One is organised long on pollutant and wide on test type; the other is wide on pollutant beneath a two-level header, with family attributes distributed on both sides of the measurement block. Both place their true column headers on the second row, so the naive read of either file produces plausible but incorrect data without raising an error.

This report documents CIDEX, a pipeline that reconciles both sources into a single analysis-ready panel of **8,627 engine families**, **9,794 engine configurations** and **94,869 emission records** across **108 manufacturers** and model years 2011–2027. It also documents a resolved carryover lineage linking **2,541 distinct certification families** across model years — a structure present in the nonroad source but not, as far as the author is aware, previously published in resolved form.

The report states the method in enough detail to be re-implemented independently, records each design decision with its justification, traces a single engine family through every stage of the pipeline, and — in Section 7 — documents the defects found during development and how each was detected. That last section is included because a validation record reporting no defects is not evidence of a correct pipeline; it is evidence of an incurious review.

Document. Technical report, part of the [FACET](https://github.com/osariemenimafidon/facet) research program. Repository <https://github.com/osariemenimafidon/cidex>. Licence: data and documentation CC BY 4.0; code MIT. Every quantity in this report is interpolated from the pipeline’s own `stats.json`; the appendices are imported from the pipeline modules at generation time.

1. Introduction

1.1 What certification data is

Before an engine may be sold in the United States it must be covered by a certificate of conformity issued by EPA. The unit of certification is not an engine model but an **engine family** — a group of engines expected to share emission characteristics, defined by the manufacturer and accepted by EPA. A family is tested at one or more configurations, the results are compared against the standard applicable to its service class and model year, and a certificate is issued for that family and model year.

The resulting record is the only comprehensive public account of what engines were approved to emit, by whom, and under which standard. It bears on fleet composition, technology adoption, the stringency of successive standards, and the relationship between certified and in-service behaviour.

1.2 Why it is hard to use

The obstacle is not access. EPA publishes the files openly, without authentication, and refreshes them quarterly. The obstacle is structure.

Every researcher who wants to ask a question spanning both engine categories must independently reverse-engineer two layouts, reconcile two pollutant encodings, discover that the headers are not where a spreadsheet reader looks for them, and decide what to do about several dozen edge cases. That work is duplicated across every user, documented by none of them, and performed under no obligation to be correct. Errors made in it are invisible: the failure mode is not an exception but a quietly wrong number.

CIDEX performs the reconciliation once, publicly, with provenance, and documents the decisions.

1.3 Scope and non-goals

CIDEX v1.0.0 covers the two current EPA certification files: heavy-duty highway gasoline and diesel engines from model year 2015, and nonroad compression-ignition engines from model year 2011. EPA’s archive files — highway 1982–2016 and nonroad 1996–2011 — are **not** ingested in this version. Their omission has a measurable effect on coverage, quantified in §8.3 and carried as a flag on every affected row.

CIDEX is explicitly **not**:

- *A compliance instrument.* It is a reorganisation of a published record, not an authoritative statement of any engine’s regulatory status. EPA’s certificate is authoritative; this is a convenience.
- *A correction of EPA’s data.* Where the source contains values that appear wrong, CIDEX publishes them as published and flags them. §5.4 explains why.
- *A source of in-service emissions.* Certification results are laboratory values on a certification fuel and duty cycle. They are not what an engine emits in a field.

1.4 What this report is for

It should be possible to re-implement CIDEX from this report alone and obtain the same numbers. Where a choice was made that a competent person could reasonably have made differently, the choice is stated as a choice rather than presented as the only option.

2. Sources

2.1 The two workbooks

	Highway	Nonroad
Model years	2015–2026	2011–2027
Source rows	7,804	10,214
Unique families	702	7,925
Manufacturers	27	86

The highway workbook carries three sheets — **Family Info**, **Model Info** and a 630,000-row **Parts Info**. CIDEX v1.0.0 uses **Family Info** only; the other two are noted as candidate extensions in §10. The nonroad workbook carries **Family Info** and a **Model Info** sheet of 80,719 rows, likewise unused.

Both files are US Government works and carry no copyright restriction on reuse.

2.2 The access constraint

The build environment used for this work cannot reach `epa.gov`: outbound requests are refused by an organisational egress policy at the network layer. The source files were therefore downloaded manually through an ordinary browser and placed in `data/raw/`.

This is a genuine weakness. An automated fetch is verifiable in a way a manual one is not: it records what was requested and what came back, and it can be repeated. A manual download records nothing, and a file placed in a directory by hand could be anything.

It is mitigated rather than concealed. `scripts/01_provenance.py` refuses to run if either expected file is absent, and records for each one its name, byte count, SHA-256 digest, filesystem modification time, and the canonical URL it is asserted to have come from. A rebuilder need not accept the assertion: they fetch the file themselves and compare digests. If the digests match, the bytes are identical and the provenance question is settled regardless of how either copy arrived. The complete record is Appendix A.

2.3 Source volatility

EPA refreshes these files quarterly, replacing them in place at the same URLs. There is no versioned archive of previous drops and no changelog. A rebuild against a later drop will therefore produce different numbers, and the digests in Appendix A will not match.

This is a property of the source, not a defect in the pipeline, but it has a practical consequence: **the digest, not the URL, identifies the data**. A citation of CIDEX should reference the deposited version, which carries its own fixed digests, rather than the live EPA file.

3. The harmonization problem

3.1 The headers are on the second row

Both workbooks reserve row 1 for a merged group header and place the actual column names on row 2. `pandas.read_excel` defaults to `header=0`.

Applied to either file that default returns a frame whose columns are named `Unnamed: 0` through `Unnamed: 44`, and whose first data row contains the real header strings. No exception is raised. Every subsequent selection by column name then either fails loudly — the good case — or, if the analyst falls back to positional indexing, succeeds against a frame whose first row is not data.

CIDEX reads the highway sheet with `header=1`. The nonroad sheet is read with `header=None` and both header rows are reconstructed explicitly, for the reason in §3.2.

3.2 The files are organised in opposite directions

The **highway** file is long on pollutant. One row carries a single pollutant for a single family and configuration, identified by a **Pollutant Name** column taking 10 distinct values. The same row is simultaneously *wide* on test type: transient and steady-state results occupy separate column blocks side by side.

The **nonroad** file is the reverse. One row carries every pollutant as a separate column, arranged in four groups beneath a two-level header:

- **Certification Level Steady-State Discrete Modal** — pollutant columns
- **Certification Level Transient Test Results** — pollutant columns
- **FEL** — three limit columns
- **Smoke Opacity** — three opacity measures

Family attributes appear both *before* the measurement block (columns 0–16) and *after* it (columns 39–45), which defeats any approach that assumes attributes and measurements occupy contiguous regions.

Reconciling these two orientations into one schema is the substance of the pipeline. The target is a single long form keyed on (**panel, model year, engine family, pollutant, test type**).

3.3 The engine family name is a true key

In both panels, the count of distinct engine family names equals exactly the count of distinct (model year, family) pairs — 702 and 7,925 respectively.

This is not a coincidence. EPA engine family names encode the model year in their first character, so a family name is unique across years by construction. TCPXL18.1NYS and CCPXL18.1NYS differ in the first character and are the same underlying family in different years.

The practical consequence is that no surrogate key is needed and a name collision across model years is impossible. CIDEX records the family name as the key and does not synthesise an identifier, which keeps every row traceable to EPA’s own record by a value a reader can look up directly.

3.4 Families and configurations are different things

A single engine family may be certified across several configurations. In the nonroad source, **1,487 of 7,925 families occupy more than one row** — up to **34 rows for a single family** — differing by engine code, engine model, displacement, certification fuel, engine operation or test procedure.

Placing those attributes in a family-level table would force an arbitrary row to win and silently discard the rest. CIDEX separates them:

- attributes **constant** within a family go to the family table;
- attributes that **vary** within a family go to a configuration table.

The split is not asserted. The build **tests** it and aborts if any attribute remaining in the family table varies within a family. Before the split, 7 nonroad attributes varied (**engine_model, engine_code, displacement_l, certification_fuel, engine_operation, test_procedure, test_procedure_type**); after it, 0 do.

The highway panel needed no such correction: 0 of its family attributes varied within a family, because its configuration-level columns were routed to the configuration table from the outset.

4. Method

4.1 Pipeline

Nine numbered scripts, each independently re-runnable, run in order:

Script	Function
01_provenance.py	Hash and log every raw file
02_harmonize_highway.py	Highway source → CIDEX schema
03_harmonize_nonroad.py	Nonroad source → CIDEX schema
04_carryover.py	Resolve certification lineage
05_combine_qa.py	Combine, flag, check, write <code>stats.json</code>
06_docs.py	Regenerate the documentation set
07_publish_gate.py	Pre-publication scan
08_figures.py	Figures
09_release.py	Build the deposit archive
11_report.py	Generate this report

4.2 Highway harmonization

The sheet is read with `header=1`. Family attributes are projected into the family table and configuration attributes — engine code, engine test model, engine identifier, test fuel, test date — into the configuration table, yielding 875 configurations against 702 families.

The pollutant name is mapped through a controlled vocabulary (Appendix C). **If any source value fails to map, the script raises.** It does not drop the row. A silently discarded pollutant is precisely the class of error this pipeline exists to prevent, and a pipeline that quietly ignores what it does not understand is worse than one that stops. In this build 0 values failed to map.

The test-type block is then unpivoted. Each source row yields one record per test type. Records with no value in any measurement column are removed, as they are artefacts of the unpivot rather than facts about an engine: 15,608 candidate rows, 2,184 dropped, **13,424 retained**.

4.3 Nonroad harmonization

The two-level header is resolved without hard-coding column positions, because positional assumptions break the moment EPA inserts a column.

The group row is forward-filled across the measurement block, then a column is classified as a measurement **only if** its row-2 sub-name is one the group actually contains. This test is what keeps `Engine Model` — which sits immediately after the FEL block and so inherits the forward-filled label `FEL (g/kW-hr)` — classified as an attribute rather than as a pollutant. A pure forward-fill would misclassify all seven trailing attribute columns.

The classifier found **22 measurement columns** and **24 attribute columns**, with 0 attributes unmapped.

Wide-to-long expansion produces 224,708 candidate records, of which 141,657 are empty cells in a sparse layout — most families report only a few of the 8 possible pollutants — leaving **81,445**.

Family Emission Limits are a regulatory limit, not a measured result, and are moved to their own column. **The merge is keyed on source-row identity, not on the family key.** §7.1 explains what happened when it was not.

4.4 Carryover lineage resolution

5,982 nonroad families name a predecessor family in a `Carryover Engine Family Name` column. These references chain, so a family’s certification history can be followed backwards across model years. The highway panel has no equivalent field.

Resolution is a graph traversal from each family to its root. Three cases are handled explicitly rather than by exception:

Self-reference. 30 families name *themselves* as their own carryover. EPA appears to use this to mark an

unchanged re-certification. A naive traversal reads it as a cycle of length one and either loops forever or reports a false positive; the first version of this pipeline reported 109 such “cycles”. CIDEX treats `parent(x) == x` as a terminus.

Orphans. 568 families name a predecessor that is absent from this panel — in almost all cases a pre-2011 family living in EPA’s archive file. The lineage is truncated rather than wrong, but it means **a recorded depth of 0 does not always mean “first of its line”**. Both counts are published so a user can distinguish the cases.

True cycles. Detected with a visited set. Count: **0**. The graph is a forest.

The resolved structure comprises **2,541 distinct lineages**, mean depth 1.382, maximum **14 model years**. The deepest runs `CCPXL18.1NYS → TCPXL18.1NYS`.

5. Design decisions

Five decisions shape the published data. Each could reasonably have gone the other way.

5.1 Units are never silently converted

The two panels certify in different units: highway in **g/bhp-hr**, nonroad in **g/kW-hr**, and nonroad smoke opacity in **percent**. Every record carries its units in a column and **no conversion is performed**.

The alternative — normalising everything to one system — would make the panel look tidier and would be a trap. A user aggregating across panels would obtain a plausible, wrong number with nothing to indicate anything was amiss. Leaving the units divergent makes the incompatibility visible at the point of use.

Conversion is available but must be invoked deliberately:

```
from cidex.units import to_g_per_kWh
to_g_per_kWh(0.20, "g/bhp-hr")    # 0.2682
```

`cidex.units.headroom()` raises `MixedUnitsError` rather than computing a margin across unit systems. The constant is $1 \text{ hp} = 0.7456998716 \text{ kW}$.

5.2 NMHCE is kept distinct from NMHC

Non-Methane Hydrocarbon Equivalent appears on 13 records against 14,219 for NMHC. It is a different regulatory construct, used for certain alternative-fuel engines.

Folding it into NMHC would tidy the vocabulary and destroy information that cannot be reconstructed downstream. Keeping it separate costs a user one filter expression. The general principle applied throughout: **never destroy information the user can collapse themselves**.

5.3 Nonroad standard is left null

The nonroad source states an applicable compliance standard at *family* level, as a text string, not per pollutant and test type. Inferring which standard applies to which pollutant would require encoding the tier structure and would be a guess dressed as data.

CIDEX leaves **standard** null for every nonroad record. Null means “not stated by the source”, never zero. **adj_result** and **fcl** are likewise highway-only.

5.4 Physically impossible values are published, flagged

50 records — 0.0527% of the panel, across 18 families, all nonroad — carry a **negative** certification result. An emission rate cannot be physically negative.

CIDEX does not correct, clip, or drop them. The published value stands as EPA reports it and carries `quality_flag = negative_cert_result`.

The reasoning: CIDEX's claim is to faithfully reorganise a published record. Silently repairing the source would break that claim and would hide a real finding — either EPA has a data-entry issue worth raising with them, or there is an encoding convention neither party has documented. Both are more useful to a reader than a quietly cleaned column. Users needing physically plausible values filter on one flag.

5.5 Incomplete model years are flagged in the data, not just documented

Both panels thin out at their edges for two unrelated reasons, and a time series crossing those years reads the artefact as a trend. Rather than document this only in prose, every row of the family and emission tables carries `year_coverage`. §8.3 quantifies it.

6. Validation

Validation operates at three levels: mechanical checks that run on every build, an evidence test for a claim that would otherwise be an assertion, and a human gate that machines cannot substitute for.

6.1 Integrity checks

10 checks run on every build; all currently pass. They test **internal consistency** and cannot establish that any value matches EPA's record.

1. `family_key_unique` — PASS

No (panel, model year, engine family) triple appears twice in the family table. The family key would otherwise be unusable for joining.

2. `no_emission_row_without_family` — PASS

Every emission record joins a family row. An orphan measurement would indicate the two tables were built from different row sets.

3. `units_never_null` — PASS

The units column is populated on every emission record, so no value can be read without its unit system attached.

4. `no_mixed_units_within_panel` — PASS

No panel carries more unit systems than it should — one for highway, two for nonroad, the second being smoke opacity.

5. `highway_units_evidenced_by_standards` — PASS

The modal certification standard per pollutant in the highway panel matches the known US heavy-duty standards expressed in g/bhp-hr. See §6.2.

6. `highway_units_labelled` — PASS

Every highway record carries the g/bhp-hr label.

7. `nonroad_units_correct` — PASS

Nonroad records carry only g/kW-hr or percent opacity, never a highway unit.

8. `carryover_no_true_cycles` — PASS

The lineage graph contains no cycle, so every traversal terminates and every family has a well-defined root.

9. `every_negative_value_is_flagged` — PASS

Every negative certification result carries a quality flag. An unflagged one would enter an unsuspecting user's aggregate.

10. `flagged_share_below_0pt1pct` — PASS

Flagged records remain under 0.1% of the panel. A sharp rise would indicate a parsing regression rather than a change in EPA's data.

6.2 The units evidence test

The highway source file states its units **nowhere**. Its column headers are bare — **TR Cert Result, Transient Standard**. The claim that highway results are in g/bhp-hr was originally an assertion written into the codebook by inference.

It is now a test. US heavy-duty highway standards have known values in g/bhp-hr, and the modal certification standard per pollutant is compared against them:

Pollutant	Known standard	Modal value in CIDEX
NOx	0.2	0.2
PM	0.01	0.01
NMHC	0.14	0.14
CO	15.5	15.5

Expressed in g/kW-hr the same standards would be 0.268, 0.013, 0.188, 20.786 — an order of magnitude away in every case. The test therefore cannot pass by coincidence, and it runs on every build.

The distinction matters beyond this one field. The inference was correct, but inference was the wrong instrument: the same reasoning applied in a sibling project substituted a European fuel specification for a US regulatory one and produced a materially wrong result.

6.3 The author verification gate

No mechanical check can establish correspondence with the source. That requires a person to look up specific records in EPA's own certificate tool and compare them by eye. The pipeline ships five named spot-checks, each probing a different failure mode:

Check 1 — highway, `MSZXH05.23FD`, MY2021

CIDEX records NOx on the transient cycle as **0.17 g/bhp-hr**. Highway, transient NOx — the standard case.

Check 2 — highway, `FNGCH0466AEA`, MY2015

CIDEX records PM on the steady state cycle as **0.01 g/bhp-hr**. Highway, earliest model year present — tests the MY2015-16 archive boundary.

Check 3 — nonroad, `LCEXL60.0AAB`, MY2020

CIDEX records NOx on the steady state cycle as **0.48 g/kW-hr**. Nonroad, steady-state NOx — tests the wide-to-long unpivot.

Check 4 — nonroad, `TCPXL32.1NZS`, MY2026

CIDEX records NMHC on the steady state cycle as **0.08 g/kW-hr**. Deepest carryover lineage (depth 14, root `CCPXL32.0NZS`) — tests lineage resolution.

Check 5 — nonroad, `CDICL05.8HTA`, MY2012

CIDEX records NOx on the steady state cycle as **-1.0 g/kW-hr**. A flagged negative value — is this real in EPA's own record?.

Publication is blocked until the author signs `docs/VERIFICATION_CHECKLIST.md`. `scripts/09_release.py` refuses to build a deposit archive while the attestation file is absent, and every document and figure regenerates stamped as unverified without it.

The attestation **pins which records were checked**. The spot-check selector chooses a median row per category, so its output moves whenever the underlying row set changes — as it did when the defect in §7.1 was corrected. Without pinning, a signature could come to refer to records nobody had ever examined. `scripts/07_publish_gate.py` compares the pinned list against the current build and blocks if they differ.

Status: Signed by Imafidon, Osariemen (ORCID 0009-0006-3069-4674) on 2026-09-15.

7. Defects found and corrected

Three defects were found during development. They are recorded because a validation section reporting none says more about the thoroughness of the review than about the correctness of the pipeline.

7.1 The FEL merge multiplied rows

Severity: data corruption. Detected: adversarial review immediately before publication.

Family Emission Limits were merged onto the emission records using the key (`model_year`, `engine_family`, `pollutant`). Because 1,487 families occupy more than one source row, that key matched several FEL entries per family, and a left join against a non-unique key fans the left side out to match.

The result was **2,139 phantom rows** — genuine values, duplicated. This is worse than missing data. No nulls appeared, no outliers, nothing downstream looked wrong; every count was simply about 2.6% too large, in a direction and pattern that looked entirely plausible.

It was found by asking a question with an arithmetic answer: how many emission rows *should* there be? The wide-to-long expansion produces a known number of candidates, of which a known number are empty and a known number are FEL entries. The remainder is determined. It did not match.

Corrected by carrying source-row identity through the wide-to-long expansion and merging on it. The script now aborts if the row count changes across the merge. Nonroad emission records fell from 83,584 to 81,445.

Generalisation: every join against a key that is not provably unique on the right-hand side is a row-multiplication defect waiting to happen, and the failure is silent. The pipeline now asserts row-count stability across the only join it performs.

7.2 The units claim was asserted, not evidenced

Severity: unverified claim in published documentation. Detected: same review.

`g/bhp-hr` appeared in the codebook as a statement of fact, having been inferred from domain knowledge rather than read from the file or tested against the data. The inference was correct. The method was not sound, and §6.2 describes the replacement.

7.3 Carryover self-references were counted as cycles

Severity: incorrect summary statistic. Detected: inspecting output that looked odd.

The first lineage traversal reported 109 cycles. Reading the list showed every one was a family naming itself — `HFPXL10.3TR3` → `HFPXL10.3TR3` — which is a terminus, not a loop. Corrected by treating self-reference as a stopping condition and detecting true cycles with a visited set. True cycles: **0**.

Generalisation: the defect was found by looking at the output rather than the summary. “109 cycles” is a number one can accept; the list of 109 pairs is not.

8. Results

8.1 What the panel contains

Table	Rows
<code>cidex_family.csv</code>	8,627
<code>cidex_config.csv</code>	9,794
<code>cidex_emissions.csv</code>	94,869
<code>cidex_carryover.csv</code>	7,925

108 manufacturers, 14 pollutants, 3 test types, units g/bhp-hr, g/kW-hr, pct opacity.

8.2 Pollutant coverage

Pollutant	Records
CH4	7,456
CO	14,238
CO2	13,196
HCHO	503
N2O	4,700
NH3	3
NMHC	14,219
NMHCE	13
NMHC_NOx	6,712
NOx	14,238
PM	14,190
smoke_accel	1,853
smoke_lug	1,763
smoke_peak	1,785

8.3 Model-year coverage

Both panels are structurally incomplete at their edges for two unrelated reasons. EPA holds older model years in separate archive files that CIDEX v1.0.0 does not ingest, so the earliest year of each panel contains only whichever families happen to sit in the current file. Separately, the newest model year is still being certified when the file is downloaded.

A year is flagged when it holds under 50% of the median count of that panel's interior years:

Panel	Model year	Families	Share of median
highway	2015	3	5%
highway	2016	29	45%
nonroad	2011	66	12%
nonroad	2027	8	2%

Usable ranges for trend analysis: highway **MY2017–2026**, nonroad **MY2012–2026**. Every affected row carries `year_coverage`.

The 50% threshold is a judgement. It separates the observed cases cleanly — no year sits near the boundary — but the raw counts are published so a reuser with a different purpose can choose a different cut.

9. A family traced end to end

To make the schema concrete, here is TCPXL18.1NYS — a nonroad family chosen because it exercises every structure this report describes.

In the source. It appears in the nonroad `Family Info` sheet with its measurements spread across the wide pollutant block, its attributes split before and after that block, and a `Carryover Engine Family Name` pointing at its predecessor.

In `cidex_family.csv`. One row, keyed (`nonroad`, 2026, TCPXL18.1NYS). Manufacturer *Caterpillar Inc.*; power category 560<kW<=2237; tier *Tier 2*.

In `cidex_emissions.csv`. 33 records — the wide row unpivoted into one row per pollutant and test type. Pollutants present: CO, CO2, NMHC, NMHC_NOx, NOx, PM, `smoke_accel`, `smoke_lug`, `smoke_peak`. Test types: `smoke`, `steady_state`. Every record carries `units` = g/kW-hr except smoke opacity, and `standard` is null for all of them, per §5.3.

In `cidex_carryover.csv`. Lineage root CCPXL18.1NYS at depth 14 — the deepest chain in the panel, meaning this family’s certification can be traced back 14 model years through successive carryovers.

A reader wanting to verify CIDEX end to end can take this family name, look it up in EPA’s interactive certificate tool, and compare every value above against the source.

10. Limitations and future work

`docs/LIMITATIONS.md` states the limitations in full. In summary: units are not comparable across panels without deliberate conversion; 50 records carry physically impossible negative values, published as EPA reports them; nonroad records have no pollutant-specific standard; edge model years are incomplete; manufacturer names are not normalised across years; and carryover lineage is nonroad-only with 568 open edges.

Candidate extensions, in rough order of value:

1. **The archive files.** Ingesting highway 1982–2016 and nonroad 1996–2011 would close the coverage gaps in §8.3 and extend the carryover lineage past its current orphan boundary.
2. **The Model Info sheets.** Both workbooks carry engine-model detail — displacement, rated power, aspiration — that CIDEX does not currently use.
3. **Manufacturer normalisation**, published as a crosswalk rather than applied silently.
4. **Annual versioning.** EPA refreshes quarterly; an annual rebuild with a diff against the previous version would turn CIDEX into a time series of the record itself.

11. Reproducibility

```
pip install -r requirements.txt
./verify.sh
```

`verify.sh` rebuilds from `data/raw/` and diffs every value against the shipped `docs/stats_reference.json`, excluding only the build date, then prints the five spot-checks for manual comparison. A mismatch is a finding.

Every quantity in every CIDEX document — this report included — is interpolated from `data/processed/stats.json`, which the pipeline writes. The column mappings in Appendix B are imported from the pipeline modules at generation time. Neither can drift from the code.

Appendix A — Provenance record

`heavy-duty-gas-and-diesel-engines-2015-present.xlsx`

- Source: <https://www.epa.gov/system/files/documents/2026-03/heavy-duty-gas-and-diesel-engines-2015-present.xlsx>
- Publisher: US EPA Office of Transportation and Air Quality
- Size: 22,908,765 bytes
- Recorded: 2026-09-15
- SHA-256:

3edfe15deef231828e26173f2a4dfe68ade0e96eb8dfe044b32cc4e63db640cf

nonroad-compression-ignition-2011-present.xlsx

- Source: <https://www.epa.gov/system/files/documents/2026-03/nonroad-compression-ignition-2011-present.xlsx>
- Publisher: US EPA Office of Transportation and Air Quality
- Size: 13,994,645 bytes
- Recorded: 2026-09-15
- SHA-256:

9e78cf00ca4f930899fed3d66101145f32dade525e8e68d5394b3df56cf5593c

Appendix B — Source-to-output column mapping

B.1 Highway family attributes

Source column	CIDEX column
Manufacturer	manufacturer
Model Year	model_year
Engine Family	engine_family
Certificate No.	certificate_no
Date Issued	date_issued
Engine Cycle	engine_cycle
Fuel Type	fuel_type
Fuel Metering System	fuel_metering_system
Introduction Date	introduction_date
Useful Life	useful_life
Intended Service Class	intended_service_class

B.2 Highway measurement blocks

Source column	CIDEX column	Test type
TR Cert Result	cert_result	transient
Transient (TR) Comb Adj Result	adj_result	transient
Transient Standard	standard	transient
Transient FEL	fel	transient
Transient FCL	fcl	transient
SS Cert Result	cert_result	steady state
Steady State (SS) Adj Result	adj_result	steady state
SS Standard	standard	steady state
SS FEL	fel	steady state
SS FCL	fcl	steady state

B.3 Nonroad attributes

Source column	CIDEX column
Model Year	model_year
Engine Family	engine_family
Manufacturer	manufacturer
Certificate #	certificate_no
Issue Date	date_issued
Commerce Introduction Date	introduction_date
Carryover Engine Family Name	carryover_family
Power Category	power_category
Applicable Regulation	regulation
Applicable Tier	tier
Applicable Compliance Standard	compliance_standard
Fuel	fuel_type
Fuel Meter System	fuel_metering_system
Useful Life of Engine Family	useful_life
Engine Combustion Cycle	engine_cycle
Non Aftertreatment Device Type	non_aftertreatment_device
Aftertreatment Device Type	aftertreatment_device
Engine Model	engine_model
Engine Code	engine_code
Displacement	displacement_l
Certification Fuel	certification_fuel
Engine Operation	engine_operation
Test Procedure	test_procedure
Test Type	test_procedure_type

Appendix C — Pollutant vocabulary

Highway source values map as below. Nonroad sub-headers map through a parallel table in `src/cidex/vocab.py`.

Source value	CIDEX code
Nitrogen Oxides	NOx
Carbon Monoxide	CO
Non-Methane Hydrocarbons	NMHC
Non-Methane Hydrocarbon Equivalent	NMHCE
Particulate Matter	PM
Carbon Dioxide	CO2
Methane	CH4
Nitrous Oxide	N2O
Formaldehyde	HCHO
Ammonia	NH3

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